

5CM510 – Network system development

Lecture 3: Peer-to-peer systems

Content

- Moving from a client-server architecture to P2P
 - Example of P2P Applications
 - Classification of P2P networks
 - Advantages
 - P2P Search Methods
 - P2P blind search
 - P2P informed search
 - Summary
-

Learning Objectives

- At the completion of this lesson you should be able to
 - Understand “cutting-edge” Internet communication technologies: P2P
 - Understand P2P data sharing and distribution

Issue

Peer to peer (P2P) is still a modern and well deployed technology, however, a large proportion of the work is now handled by networking frameworks as they make implementation a lot simpler. For C# Brunet or the Windows Communication Foundation (WCF) are excellent for this.

Modern cutting edge research tends to focus on mobile, ad-hoc, and network security issues rather than direct P2P implementations.

Information

Client-Server and P2P

- Examples of Client-Server Internet Applications:
 - World Wide Web
 - E-mail
 - Netflix
- Client-Server
 - Advantage - Efficient
 - Disadvantage –single-point-of-failure, scalability, limited resources at the server side

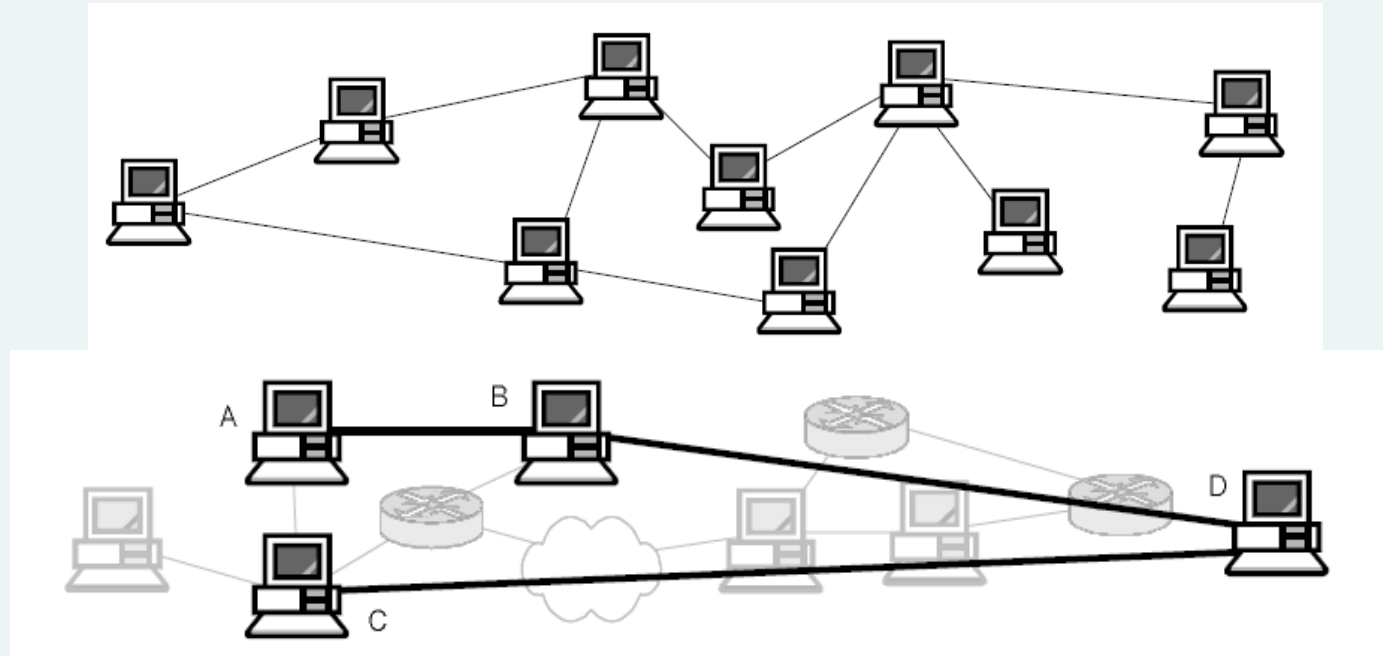
Google Data Centres

- Google spent \$10.9B in 2016 on data centres
- Each data centre uses 50-100 megawatts of power, 60B+ kWh each year (10 million UK homes)



Peer-to-Peer (P2P)

- Each peer acts as **both client and server**, requesting resources from as well as routing queries and serving resources for other peers



P2P Client Node Architecture

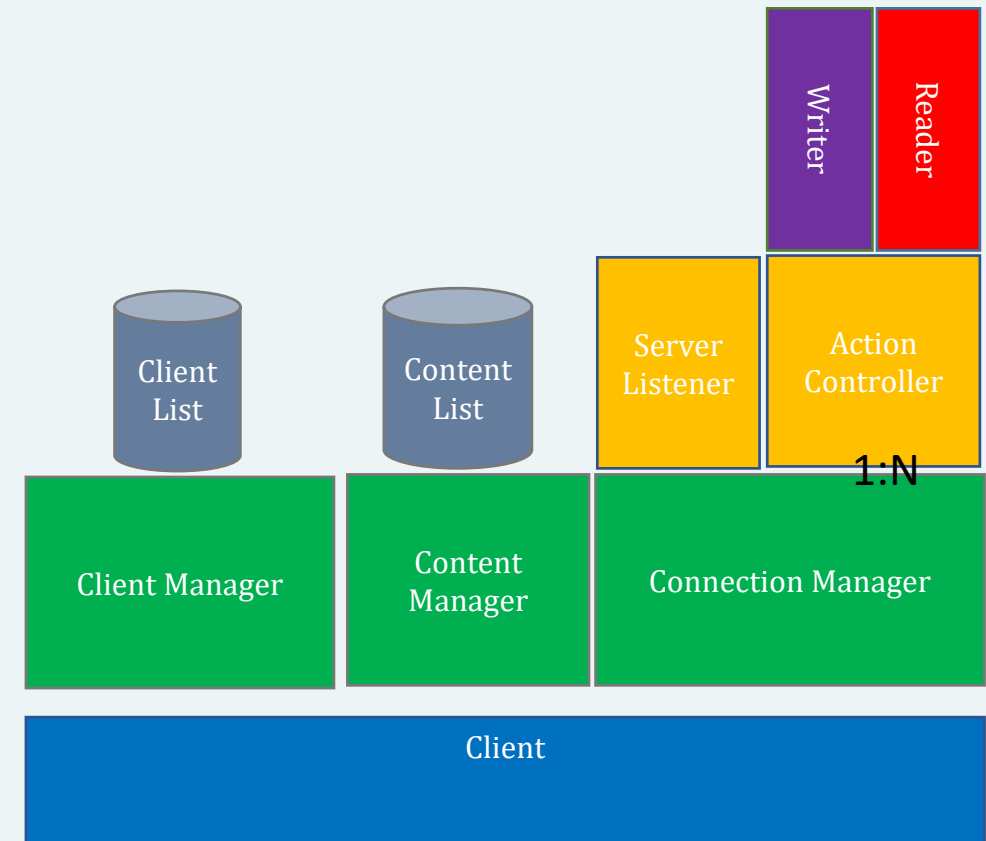
Exercise

You are tasked with writing a small torrent client with each node within the network acting as a content source, a content consumer, and a content directory.

1. What key components would you need to assemble this design?
2. How many threads per connected client do we maintain?
3. What would make this better?

P2P Client Node Architecture

- P2PNode – Main thread / GUI
 - Connection Manager
 - Server listener (new connections)
 - Action controller (existing connections)
 - Content Manager
 - Maintains a list of all known content locations
 - Client Manager
 - Maintains a list of all known clients and their behaviour



Question

- Does anyone here use any P2P software?
- Why?

- File Distribution
 - BitTorrent



“Data Networks”

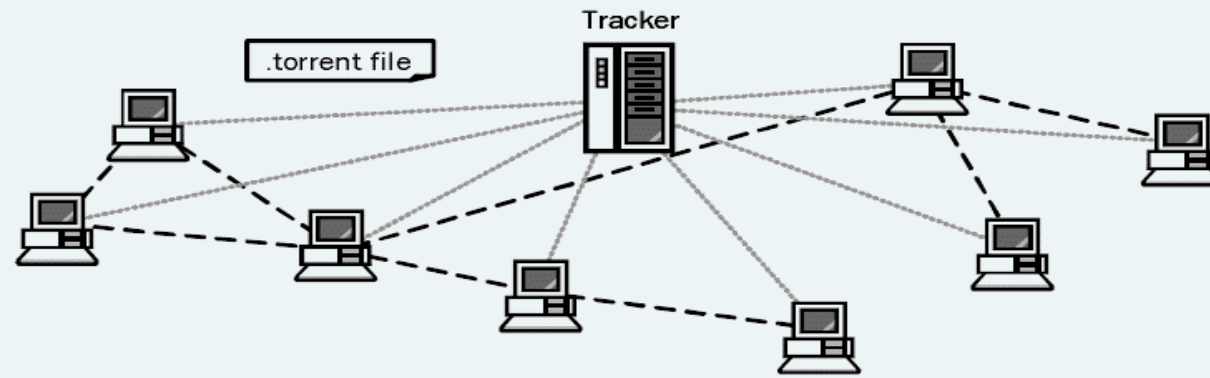
- Internet Telephony:
 - Skype



“Internet Communications”

BitTorrent

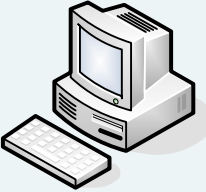
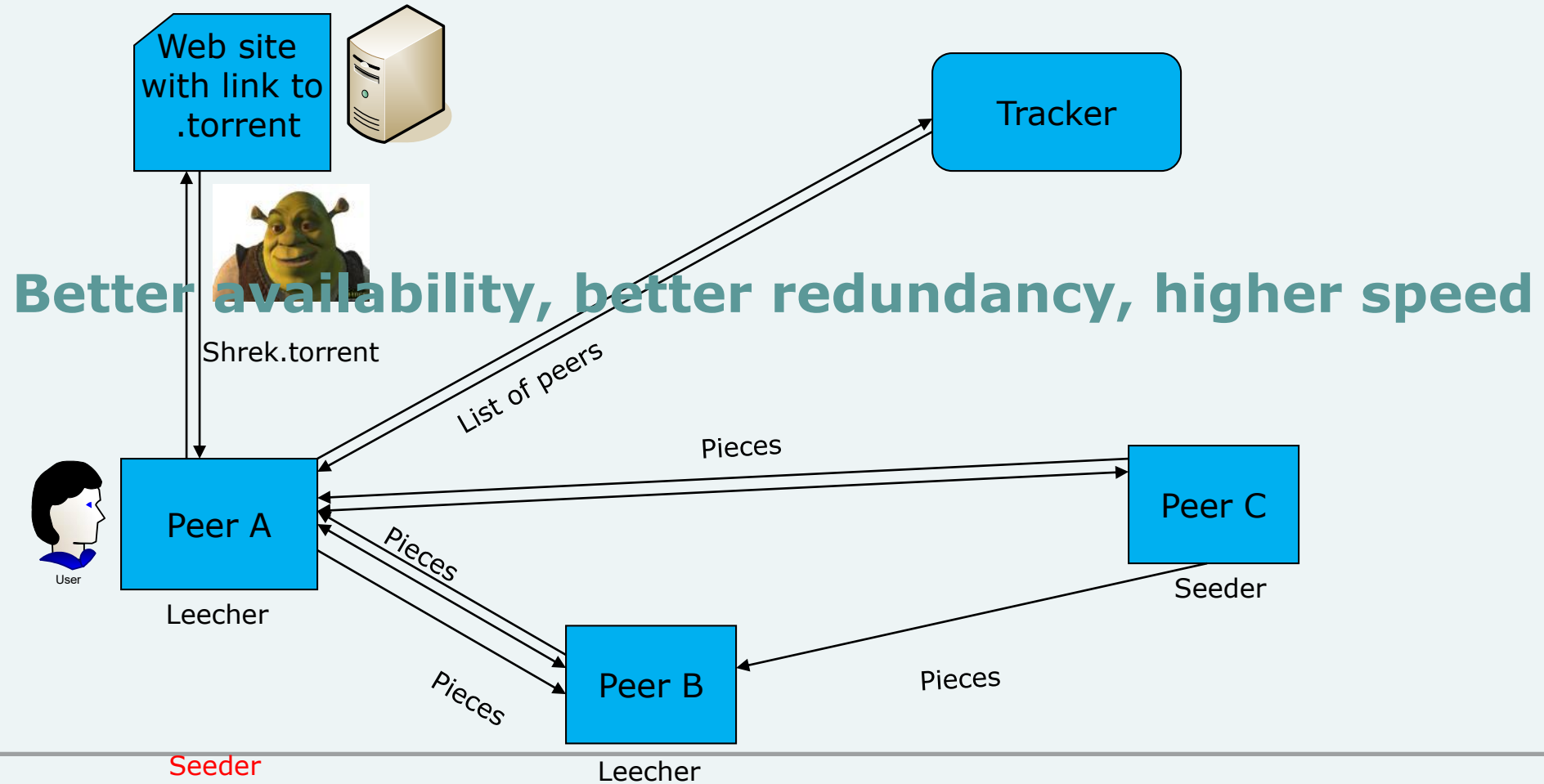
- Popular, tens of millions of users
- To distribute large amounts of data. users need to create a small .torrent file that contains the address of the tracker machine that launches the file distribution.
- **Torrent files** are located at BitTorrent download **websites**
- **Torrent files** lead downloaders to **trackers**
- **Trackers** direct downloaders to **swarms**



BitTorrent

- File is broken into **smaller fragments** (typically 256 KB each)
- Peer joining torrent:
 - has no fragments, but will accumulate them over time
 - registers with tracker to get list of peers, connects to subset of peers (“neighbors”)
 - simultaneously connects to these peer nodes that are participating in file transfer, and downloads different pieces of the file from different peer nodes.
- While **downloading**, peer **uploads** chunks to other peers.
- Terminology:
 - **Leecher**: Downloads files but does not have complete files
 - **Seeder**: Has complete files and lets others download files from him/her

How BitTorrent Works



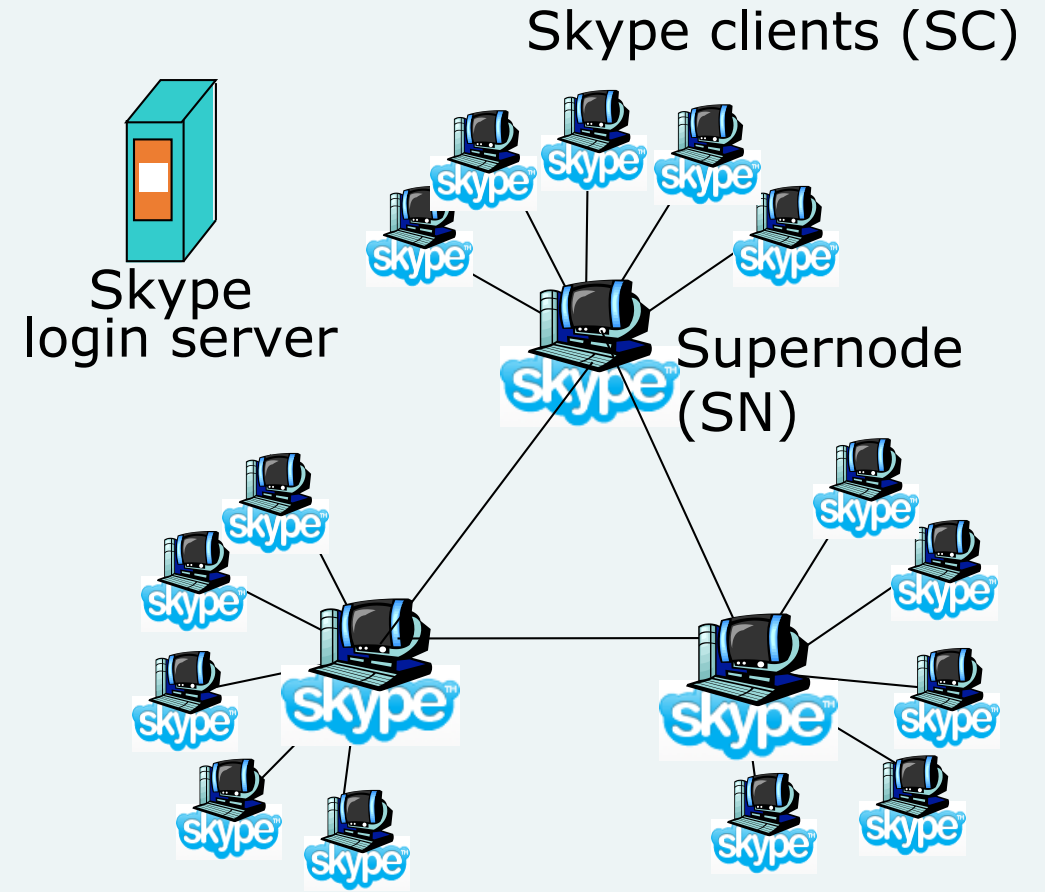
Seeder

Leecher

Seeder

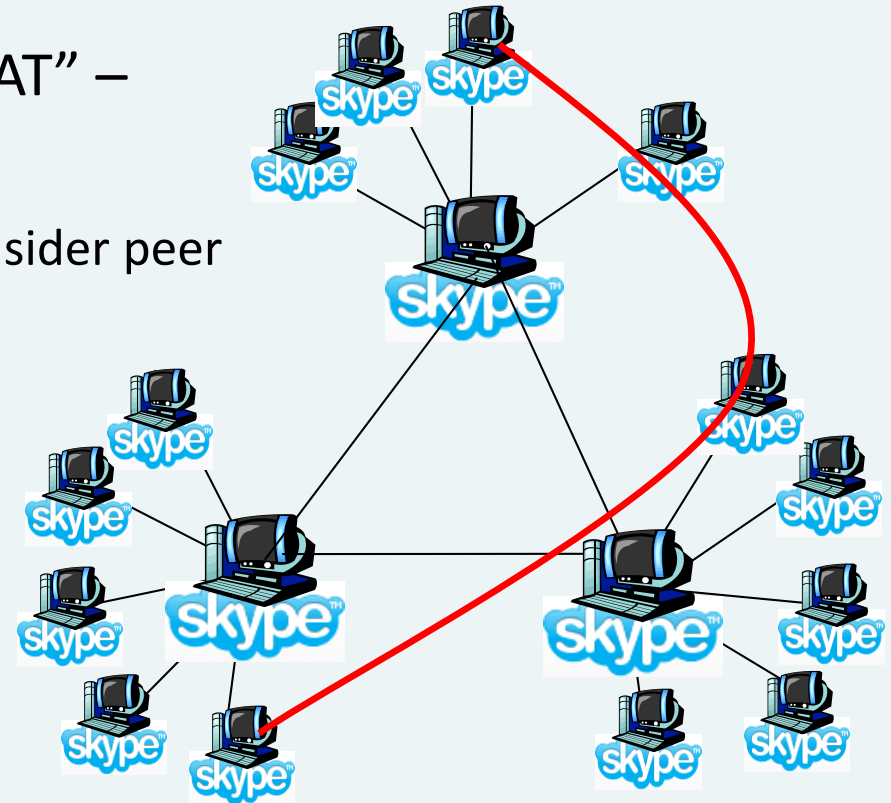
Internet Telephony - Skype

- A hybrid P2P System
 - **Client-server**: finding address of remote party
 - **P2P**: direct (not through server)
- Hierarchical overlay with Super nodes (SNs)
- Index maps usernames to IP addresses; distributed over SNs



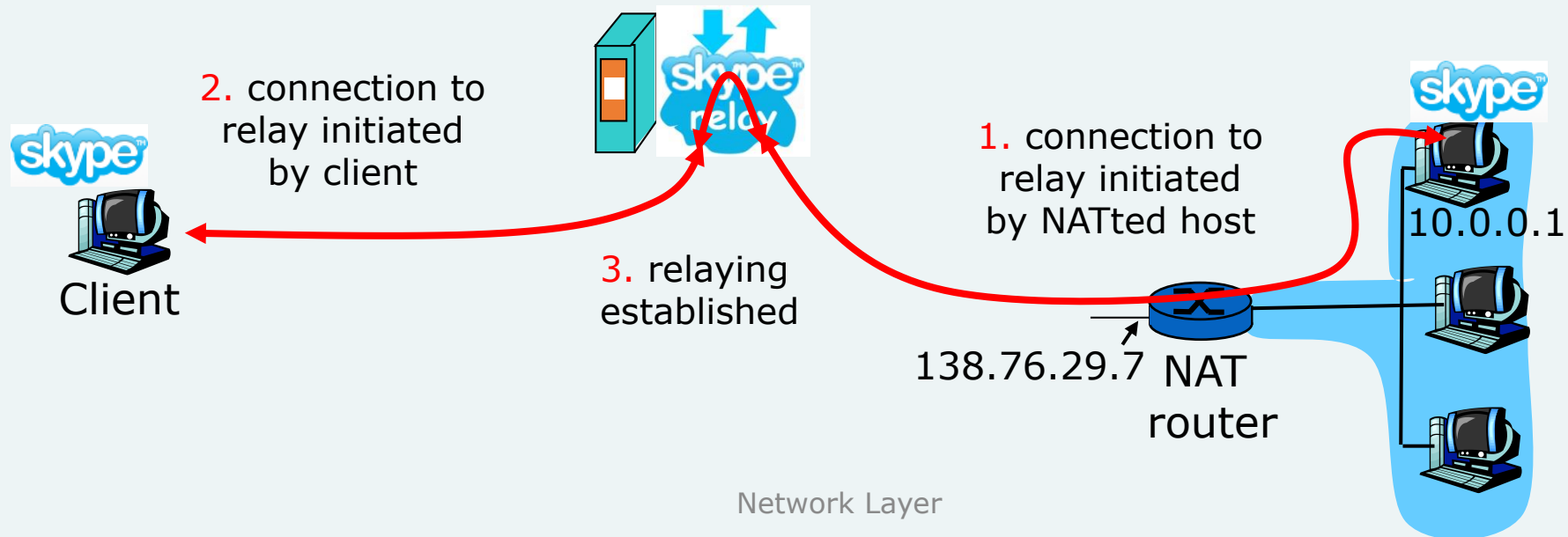
Peers as relays

- Problem when both Alice and Bob are behind “NAT” –
Network Address Translation
 - NAT prevents an outside peer from initiating a call to insider peer
- Solution:
 - Using Alice’s and Bob’s SNs, Relay is chosen
 - Each peer initiates session with relay.
 - Peers can now communicate through NATs via relay



NAT traversal

- NATed client establishes connection to relay
- External client connects to relay
- relay bridges packets between two connections

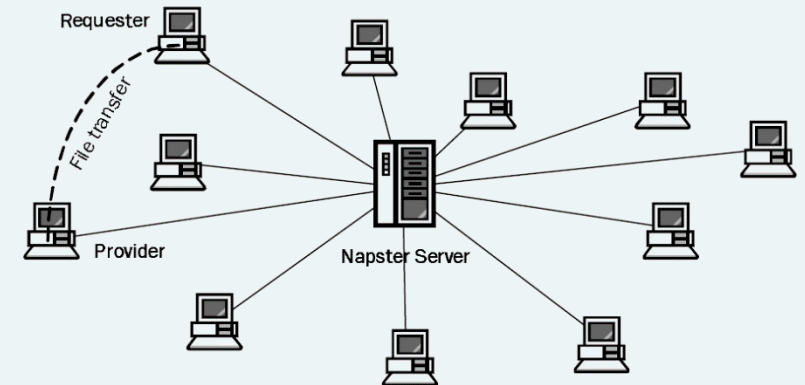


Content

- From Client Server to P2P
- Example of P2P Applications
- **Classification of P2P networks**
- Advantages
- P2P Search Methods
- P2P blind search
- P2P informed search
- Summary & next lecture

Centralised P2P Applications

- Napster
 - the first widely-used P2P music sharing service.
 - Napster utilises a simple but highly efficient mechanism to share and search files in the network.
 - To participate in the Napster network, new users need to register to the Napster server and publish a list of files they are willing to share.
 - To search a shared file in the network, users can request the Napster server and retrieve a list of providers hosting the files matching the query.
 - File transfer takes place without the Napster server participating.
 - The requested file is transferred directly between the requester and the provider



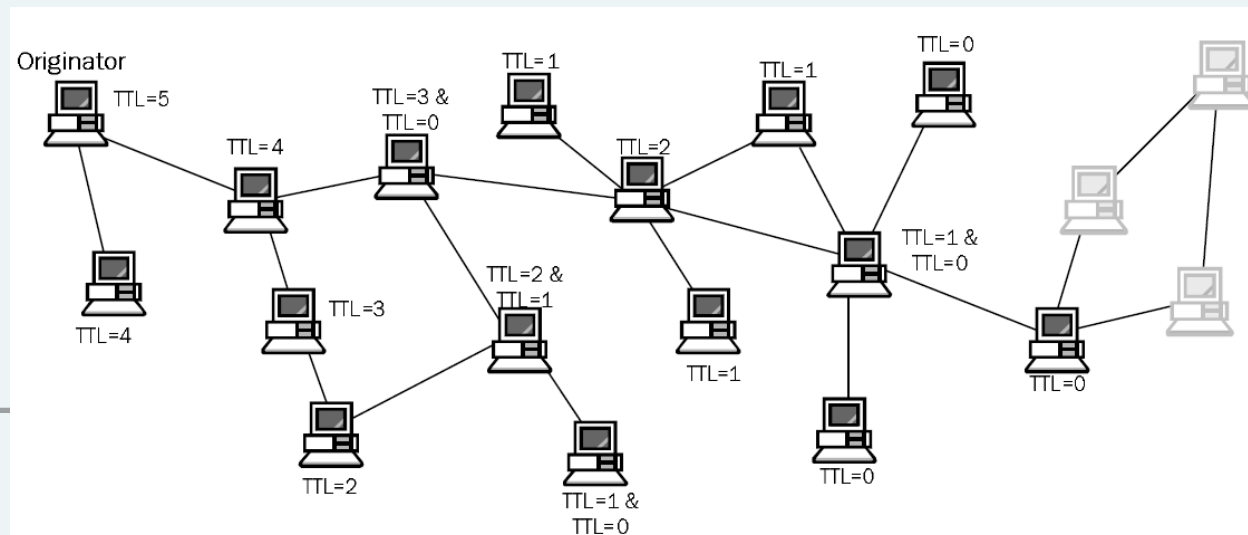
Decentralised P2P Networks

- Gnutella
 - The Gnutella network is a decentralised file-sharing P2P network, which is built on an open protocol developed to enable peer node discovery, distributed search, and file transfer.
 - Each Gnutella user needs a Gnutella client software to join the Gnutella network.
 - The Gnutella client software on the initial use can bootstrap and find a number of possibly working peer nodes in the network and try to connect to them.
 - If some attempts succeed, these working peer nodes will then become the new node's neighbours and give the new node their own lists of working nodes.
 - The new node continues to connect to these working peer nodes, until it reaches a certain quota (usually user-specified).

Decentralised P2P Networks

- Gnutella (cont.)

- Breadth-First Search (BFS) mechanism to search the network by broadcasting the query with a Time-to-Live (TTL)
- TTL represents the number of times a message can be forwarded before it is discarded.
- Each peer node receiving the query will process it, check the local file storage, and respond to the query if one matched file is found.
- Each peer node decreases TTL by one and forwards the query to all of its neighbours. This process continues until TTL is equal to zero.



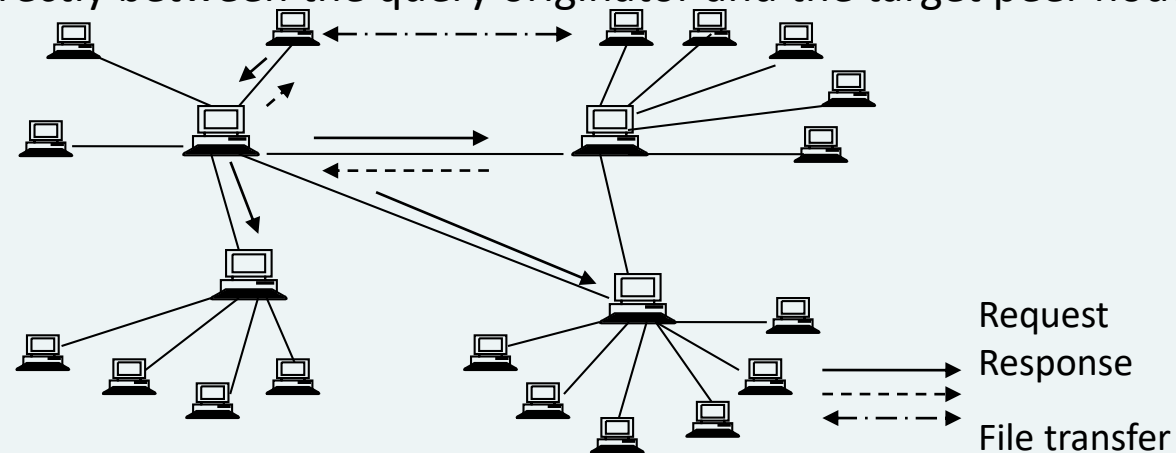
Hybrid P2P Networks

- **Kazaa**
 - Kazaa reorganises peer nodes into a two-level hierarchy with supernodes and leaves.
 - A supernode is a temporary index server for other peer nodes.
 - A newly joined node will attempt to contact an active supernode from a list of supernodes offered by Kazaa client software.
 - The newly joined node will send a list of files it shares to the connected supernodes and further retrieve more active supernodes from the connected supernodes for future connection attempts

Hybrid P2P Networks

■ Kazaa

- In Kazaa, each leave node begins a lookup by sending a lookup request to its connected supernode.
- The supernode not only checks the local index for the file requested, but also communicates with other supernodes for a list of addresses of peer nodes sharing the files.
- When a supernode discovers the requested file from its local index, it will respond to the original supernode.
- The file is transferred directly between the query originator and the target peer node that shares the file.



Advantages

- P2P frees users from the traditional dependence on central servers, which enables end users to easily share resources (e.g. music, movies, games and other software).
- P2P applications are more resilient than those built on the client-server architecture by removing the single-point-of-failure.
- P2P distributes the responsibility of providing services from centralised servers to each individual peer node in the network.
- P2P exploits available bandwidth, processor, storage and other resources across the entire network.
- P2P interactions are only between individual peers which eliminate the bottleneck of centralised servers.
- Most P2P applications use virtual channels for communication which break the obstacles of corporate private networks, such as firewalls and Network Address Translation (NAT).

Disadvantages

Exercise

In small groups discuss what possible disadvantages P2P networks have.

Describe a service which should be P2P, and one which should not be.

Disadvantages

- Security
- Trust
- Data validity and corruption
- Routing and network structure
- Network control

Question

For a P2P file-sharing application, do you agree with the statement:

“There is no notion of client and server sides of a communication session”

Why or why not?

Question

For a P2P file-sharing application, do you agree with the statement:

“There is no notion of client and server sides of a communication session”

Why or why not?

No. All communication sessions have a client side and a server side. In a P2P file-sharing application, the peer that is receiving a file is typically the client and the peer that is sending the file is typically the server.

Content

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- Example of P2P Applications
- Classification of P2P networks
- Advantages
- **P2P Search Methods**
- P2P blind search
- P2P informed search
- Summary & next lecture

Requirement

- Requirement
 - Need new mechanisms are required to compensate for the server
- Existing Methods
 - Structured P2P systems
 - Unstructured P2P systems

Peer-to-Peer Search

- Structured P2P Search
 - Establish a link between the stored content and the address of a node
 - Algorithms: distributed hash tables (Chord, ROME, Pastry, CAN, and Kademlia)
 - <https://www.cs.utexas.edu/~browne/CS395Tf2002/Papers/Keywordsearch.pdf>
 - <https://courses.cs.washington.edu/courses/cse522/05au/searchingsurvey.pdf>
 - Needs to maintain a consistent distributed index
- Unstructured P2P Search
 - Do not maintain network structure
 - Blind Search: e.g. Flooding, Iterative Deepening
 - Informed Search: e.g. Neurogrid

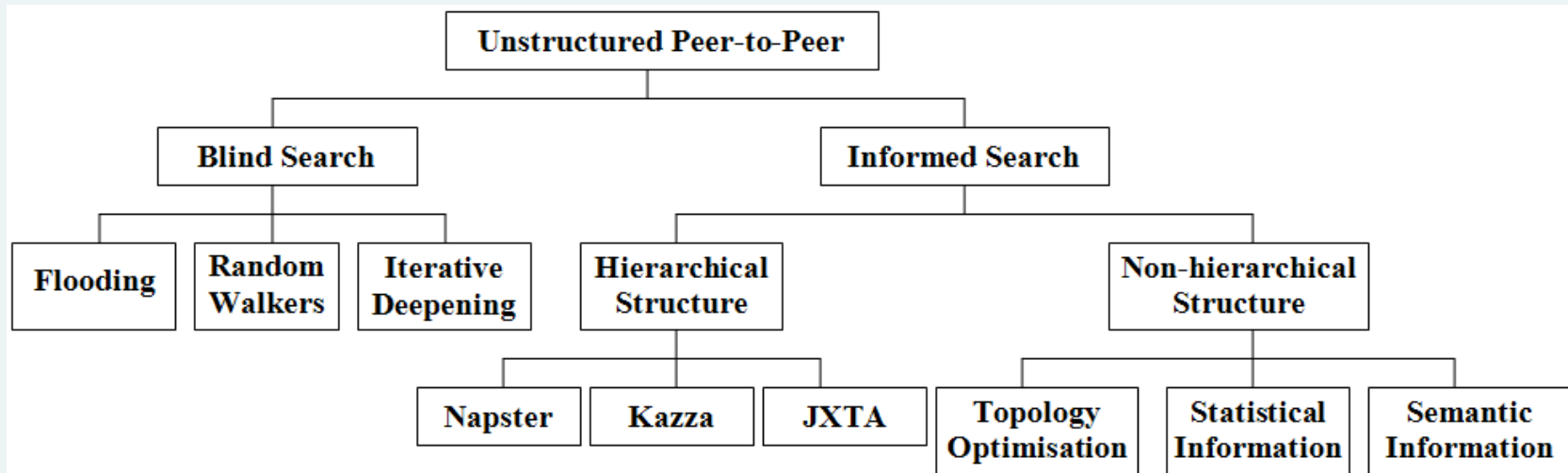
Structured P2P Systems

- Structured P2P Systems
 - Establish a link between the stored content and the IP address of a node
- Distributed Hash Table
 - Algorithms: Chord, ROME, Pastry, CAN and Kademlia
 - Each file is associated with a key generated by hashing the file name or content.
 - Each peer node in these systems is responsible for storing a certain range of keys.
 - The network structure is sorted by routing tables stored on individual peer nodes.
 - Each peer node only needs a small amount of “routing” information about other nodes.
 - Need high overhead to maintain a consistent distributed index in the dynamic Internet

Unstructured P2P Systems

- **Unstructured P2P Systems**

- E.g. Flooding, random walks, routing indices
- Do not maintain network structure
- Either require high storage overhead or generate massive network traffic to search networks

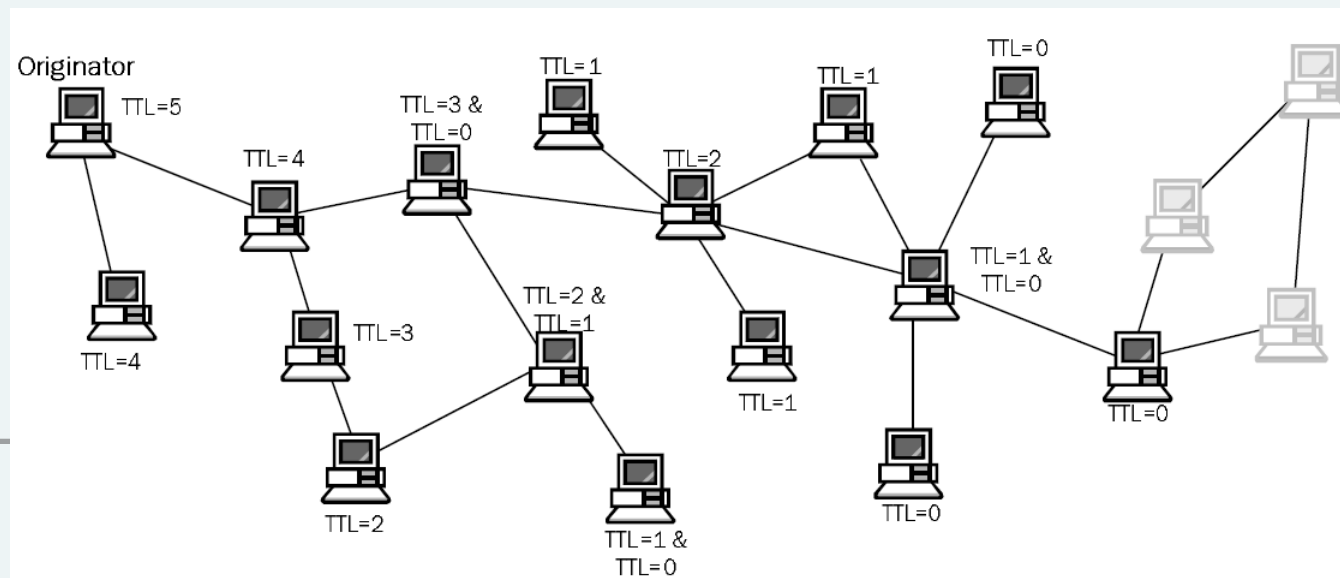


Blind Search

- Peer nodes do not maintain **additional information** about resource location.
- Advantages
 - Do not need any communication **overhead** to maintain the additional indices about resource locations
 - Extremely resilient in the highly dynamic networks.
- **Flooding** is the fundamental approach of blind search where queries are forwarded to all connected peer node to see whether they have a requested file.

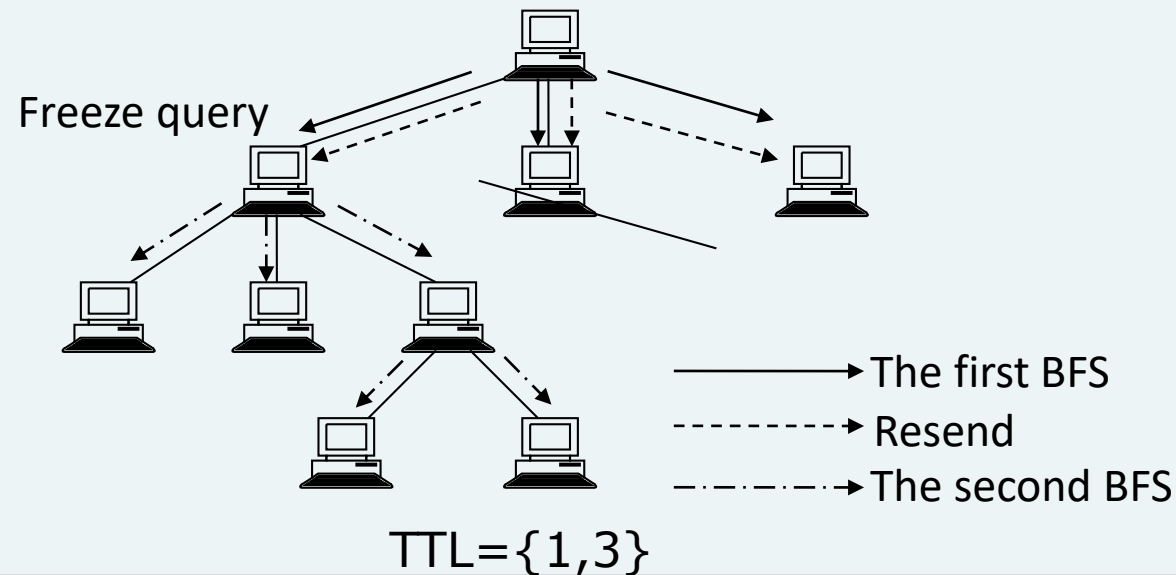
Breadth-First Search

- **Breadth-First Search (BFS)** mechanism to search the Gnutella network by broadcasting the query with a **Time-to-Live (TTL)**
- TTL represents the number of times a message can be forwarded before it is discarded.
- Each peer node receiving the query will process it, check the local file storage, and respond to the query if one matched file is found.
- Each peer node **decreases TTL by one** and forwards the query to all of its neighbours. This process continues until TTL is equal to zero.



Iterative Deepening

- **Iterative deepening** enables query originators to use **successive BFS** (Breadth-First Search) queries with **increasing depths**, until the request is satisfied or the maximum depth is reached.
- A waiting time needs to be defined between two successive BFS lookups



Informed Search

- Informed searches utilise existing information to improve search speed and quality
- A* is a classic example of informed search

<https://pdfs.semanticscholar.org/b901/ffca63868cc3093d4fffe2a3de5567d53406.pdf>

http://www.cs.cornell.edu/courses/cs4700/2011fa/lectures/03_InformedSearch.pdf

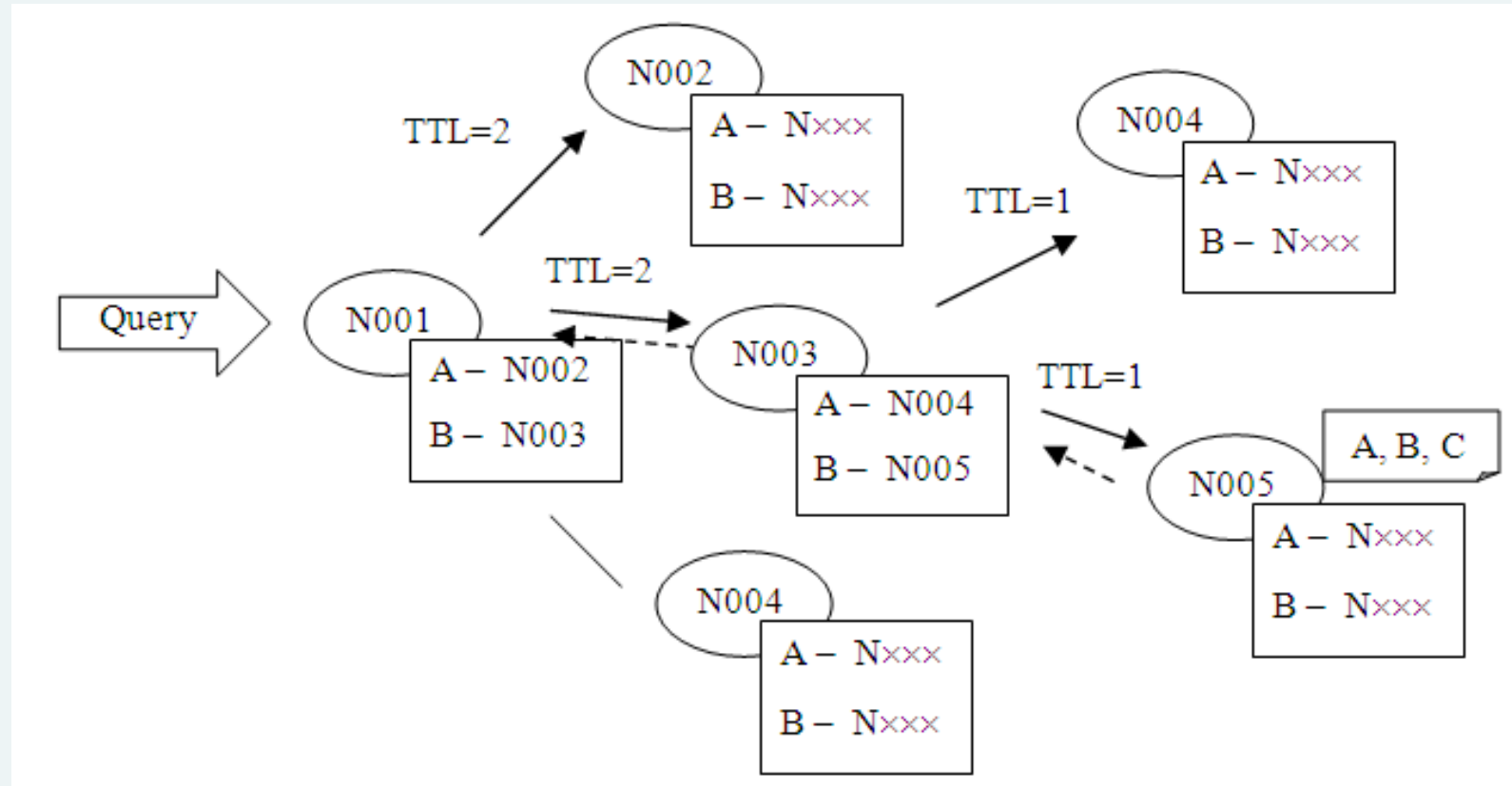
Informed Search: Neurogrid

- **NeuroGrid**

- NeuroGrid utilises the historic record of previous searches to help peer nodes make routing decisions.
- Peer nodes support distributed searches through semantic routing (keywords) by maintaining routing tables at each node.
- In the local routing tables, each peer node is associated with keywords regarding the content it stores.
- When a peer node forwards a query, it will search for the peer nodes that are associated with the query keywords.
- Users' responses to search results are stored and used to update the meta-data describing the content of remote peer nodes.
- NeuroGrid can learn the results from previous searches to make future searches more focused and semantically routes queries according to the cached knowledge.

Neurogrid Example

Keywords *A*, *B* and *C*



Structured and Unstructured

- We have discussed P2P networks in terms of their reliance on centralised nodes, however, modern classification of P2P typically talks about structure in terms of node connections and data locations

Exercise

Prepare a short presentation (no more than 6 slides) on a structured P2P network of your choice.

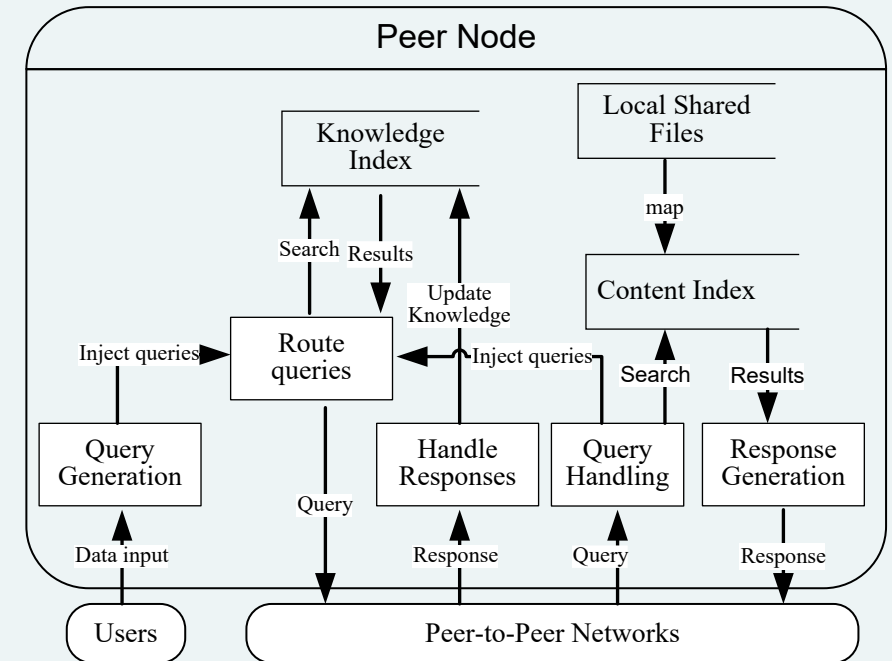
Provide examples of the technology in use, and how it works.

Provide appropriate references to your sources.

Social-Like Peer-to-peer

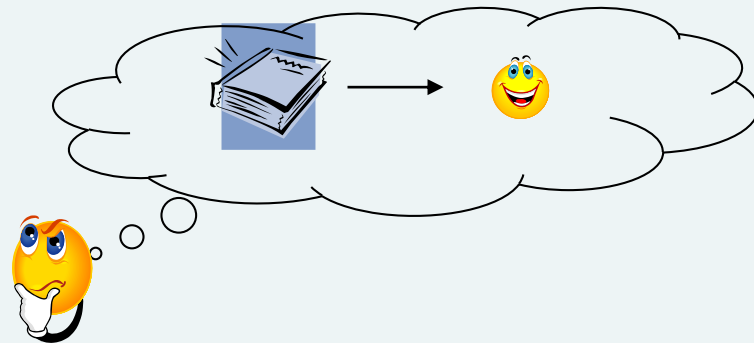
- **Motivations**

- Similarity between social networks and P2P networks
 - **Social networks** - people are connected by their social relationships
 - **P2P Networks** - two autonomous peer nodes can be connected if the users in those nodes are interested in each other's data
- Human strategies in social networks are useful for improving performance of resource discovery by **self-organising** autonomous peers on unstructured P2P networks – Efficient Social-Like Peer-to-peer (ESLP)
- Project funded by DIF DTC



Node Selection

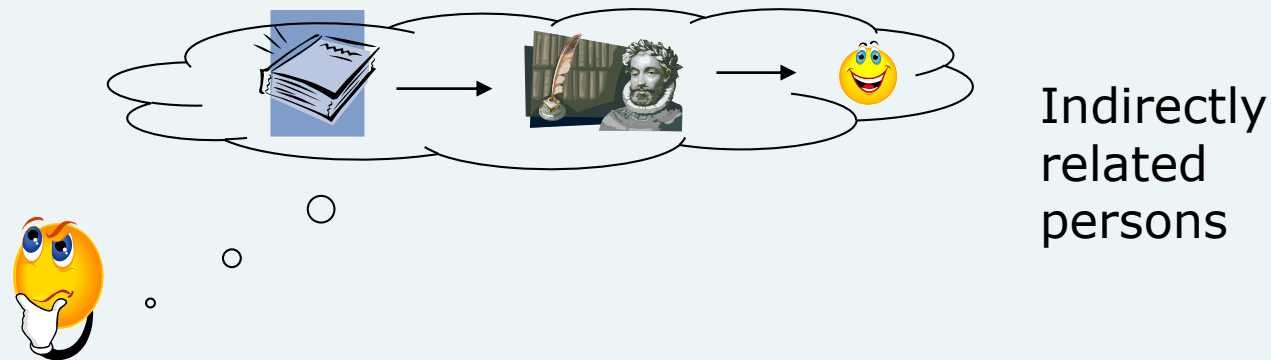
- **Social behaviour:** for resource discovery in social networks, people usually recall information in memory to find the right people to contact.
- **P2P strategy:** ESLP (Efficient Social Like Peer-to-peer) search algorithm preferentially forwards queries to peer nodes directly associated with the requested topic which have the greatest likelihood of finding the requested files.



Directly
related
persons

Node Selection

- **Social behaviour:** in most cases, people cannot find the persons who are directly related to their requests, but people can find some acquaintances that potentially have knowledge about the resources they are looking for, or can provide background information or give advice on how to find the resources.
- **P2P strategy:** Queries are forwarded to the peer nodes sharing content associated with the interest area of the requested topic in the local friend list.



Adaptive Query

- **Social behaviour:** newly joined persons normally are more active to adapt to new environments.
 - When he/she seeks out help in a new society, communication with the other people who can be of assistance is not normally limited to a strict exchange of assistance;
 - Often, he/she would like to know more about the people who can be of assistance in order to expand his/her knowledge of the new society, which may be useful at a later time.
- **P2P strategy**
 - Two kinds of queries are generated in different stages of searches, known as ordinary queries and active queries.
 - For the ordinary queries, the target nodes sharing the desired files will respond to the information related to the requested topic only.
 - For the active queries, the target nodes will not only respond to the requested topic but also inform the query originator of other associated topics it shares in the same interest area.

Adopting Social Strategies in P2P Domain

Social behaviours	ESLP patterns
1. People remember and update potentially useful knowledge from social interactions, and then random and diffuse behaviours gradually become highly organised [1].	Each peer node builds a knowledge index that stores associations between topics and the related peer nodes by the results of searches. ESLP enables peer nodes not only to forward queries to the “good” nodes that can potentially offer the requested resources or know who has the requested resources, but also to avoid sending queries to the “bad” nodes that cannot help find any requested files.
2. Some events with associated people fade from a person's memory with time [2] and a person's social network is adjustable with changing environments [3].	Peer nodes update their knowledge about other peer nodes from daily search results. Some old and invalid knowledge is replaced by newly obtained knowledge. The invalid knowledge that fails to be updated with daily searches will be dropped to the bottom of the lists and will be removed when the list reaches a maximum.
3. A social community is a group of people with common interests, goals or responsibilities which is formed spontaneously. Social communities are self-organised by daily intercommunications between people.	A node has more probability to link to other peer nodes with the same interests, which have files of interest to him/her, with a high degree of likelihood. Therefore, peer nodes that have the same interests will be highly connected to each other and form a virtual community spontaneously.
4. For resource discovery in social networks, people usually recall information in memory to find the right people to contact. The persons recalled from memory may directly relate to their requests.	ESLP search algorithm preferentially forwards queries to peer nodes directly associated with the requested topic which have the greatest likelihood of finding the requested files.
5. In most circumstances, people cannot find the persons who are directly related to their requests, but people can find some acquaintances that potentially have knowledge about the resources they are looking for, or can provide background information or give advice on how to find the resources [4].	Queries are also forwarded to the peer nodes sharing content associated with the interest area of the requested topic in the local friend list. The routing algorithm prefers to select the peer nodes with higher degrees of correlation to the interest area of the requested topic rather than the peer nodes with lower degrees of correlation.
6. Searching for a piece of information in social networks is most likely a matter of searching the social network for an expert on the topic together with a chain of personal referrals from the searcher to the expert [5].	If a peer node has a large amount of content in a particular area like an “expert”, it is very likely that it will also have other content or knowledge on this area.
7. A person does not need to tell everybody he/she is an expert in the areas which has been indicated with his/her social behaviours.	In the ESLP network, it is not necessary for a peer node to declare its interest since that has already been implied by its shared files.
8. Newly joined persons normally are more active to adapt to new environments. When a person joins to a new society, he/she will not only passively learn knowledge by remembering useful information from daily occasional events happening in the society, but also actively collect potentially useful knowledge consciously.	The recently joined nodes will utilise active queries to quickly accumulate a large amount of useful information regarding their interests. After the cached knowledge reaches a certain threshold ratio of the maximum size of the friend list in the knowledge index, the peer nodes will use ordinary queries to discover the required files with low traffic cost.

Social-Like Peer-to-peer

- **Self-adaptability**

- Each peer node has self-adaptability to modify its behaviours according to the environmental changes.
- Each node is able to self-adapt searches by using different types of queries according to its own situation to quickly adapt to changing environment and also self-justify forwarding degree according to the probability of successfully finding the requested files.

- **Self-organisation**

- Each peer node can automatically detect potential interests of other peer nodes in the network and preferentially link to the peer nodes that have similar interests.
- Global behaviours then emerge as the result of all the local behaviours that occur. Peer nodes that have the same interests will gradually connect to each other and form peer communities spontaneously.

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THANK YOU

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